

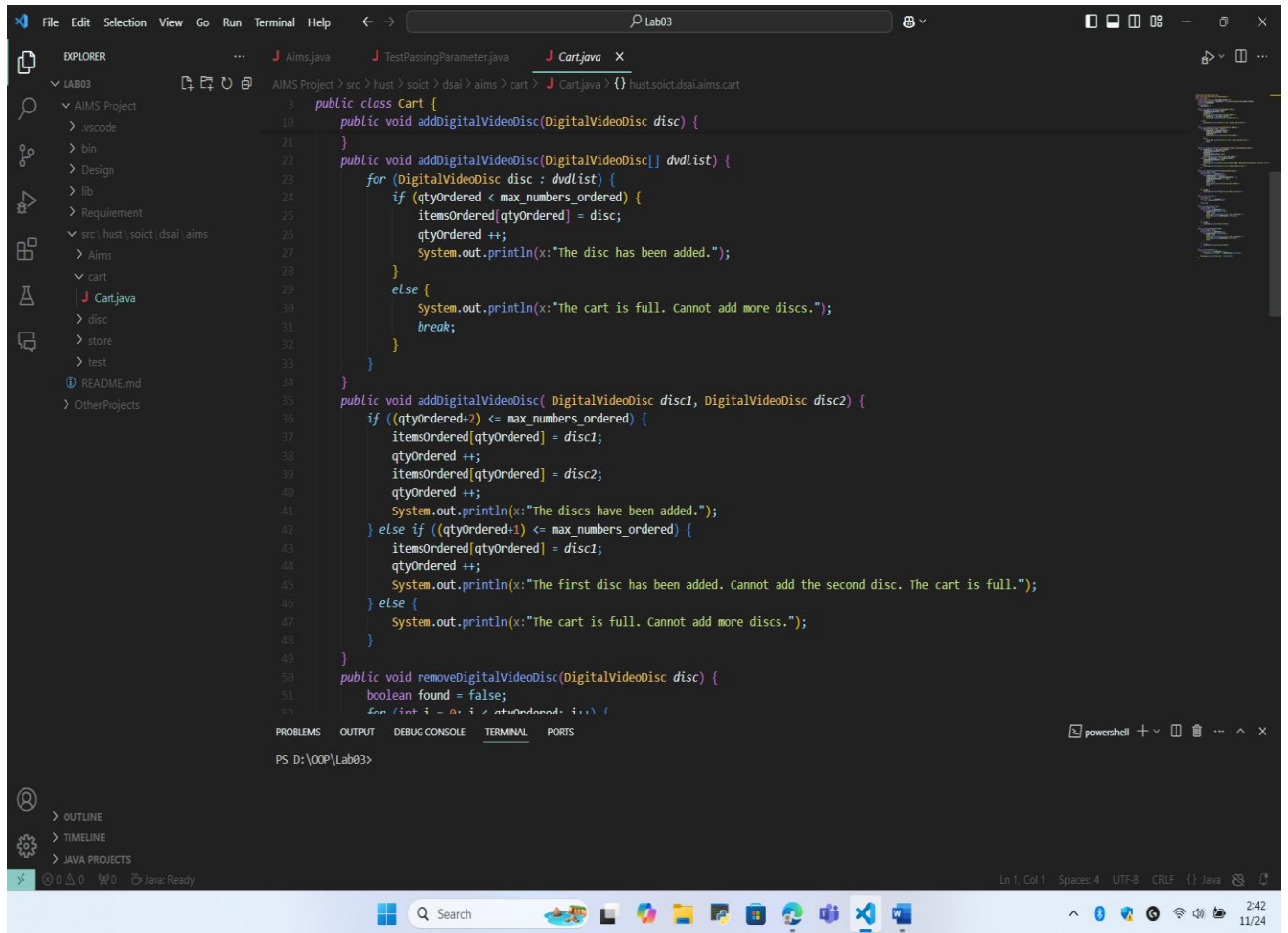
LAB03-REPORT

Name: Nguyễn Ngọc Linh

ID: 20235520

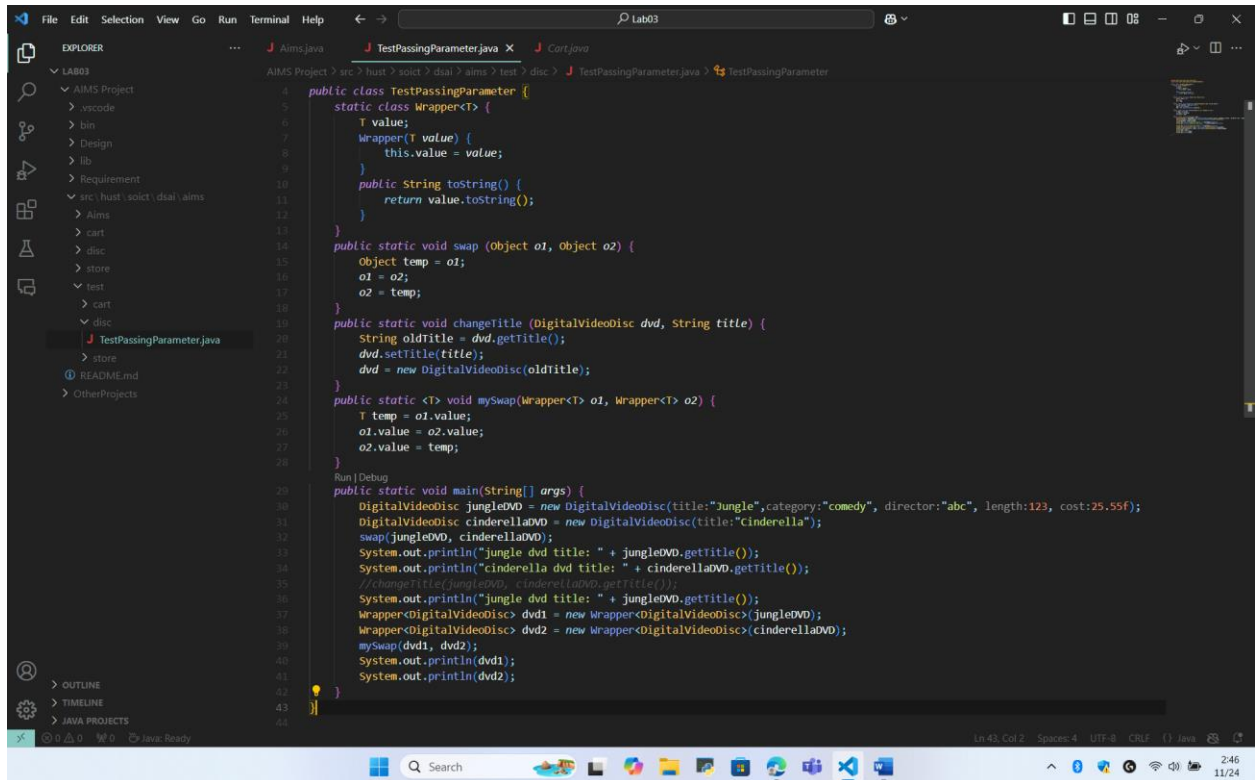
I. Source Code

1) Method overloading



```
File Edit Selection View Go Run Terminal Help Lab03
EXPLORER
  LAB03
    AIMS Project
      .vscode
      bin
      Design
      lib
      Requirement
      src\hust\solict\dsai\aims
        Aims
        cart
          Cart.java
          disc
          store
          test
        README.md
        OtherProjects
  ...
  J Aims.java
  J TestPassingParameter.java
  J Cart.java X
AIMS Project > src > hust > solict > dsa\ aims > cart > J Cart.java > {} hust\solict\dsai\aims\cart
3   public class Cart {
10  public void addDigitalVideoDisc(DigitalVideoDisc disc) {
21  }
22  public void addDigitalVideoDisc(DigitalVideoDisc[] dvdlist) {
23    for (DigitalVideoDisc disc : dvdlist) {
24      if (qtyOrdered < max_numbers_ordered) {
25        itemsOrdered[qtyOrdered] = disc;
26        qtyOrdered ++;
27        System.out.println(x:"The disc has been added.");
28      }
29    }
30    else {
31      System.out.println(x:"The cart is full. Cannot add more discs.");
32      break;
33    }
34  }
35  public void addDigitalVideoDisc( DigitalVideoDisc disc1, DigitalVideoDisc disc2) {
36    if ((qtyOrdered+2) <= max_numbers_ordered) {
37      itemsOrdered[qtyOrdered] = disc1;
38      qtyOrdered ++;
39      itemsOrdered[qtyOrdered] = disc2;
40      qtyOrdered ++;
41      System.out.println(x:"The discs have been added.");
42    } else if ((qtyOrdered+1) <= max_numbers_ordered) {
43      itemsOrdered[qtyOrdered] = disc1;
44      qtyOrdered ++;
45      System.out.println(x:"The first disc has been added. Cannot add the second disc. The cart is full.");
46    } else {
47      System.out.println(x:"The cart is full. Cannot add more discs.");
48    }
49  }
50  public void removedigitalVideoDisc(DigitalVideoDisc disc) {
51    boolean found = false;
52    for (int i = 0; i < itemsOrdered.length; i++) {
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS D:\OOP\Lab03>
OUTLINE
TIMELINE
JAVA PROJECTS
Ln 1, Col 1 - Spaces: 4 - UTF-8 - CRLF - Java 2:42 11/24
```

2) Passing parameter

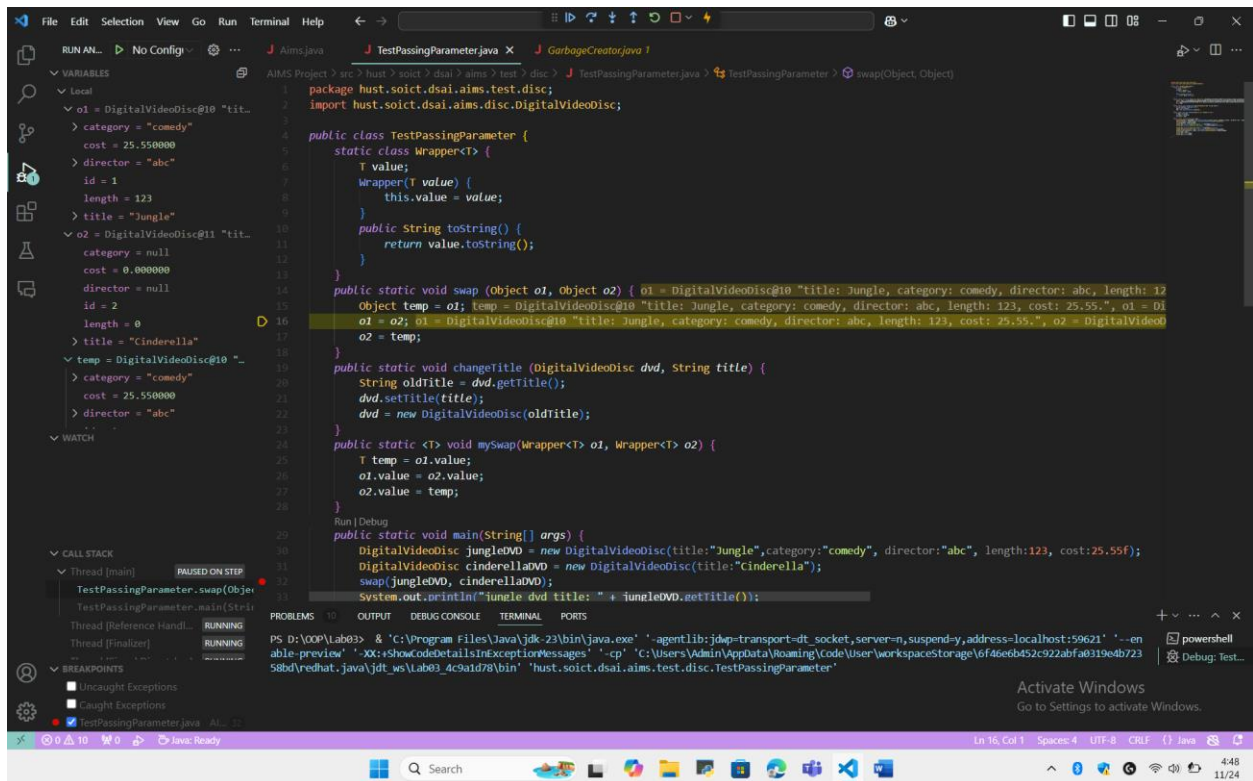
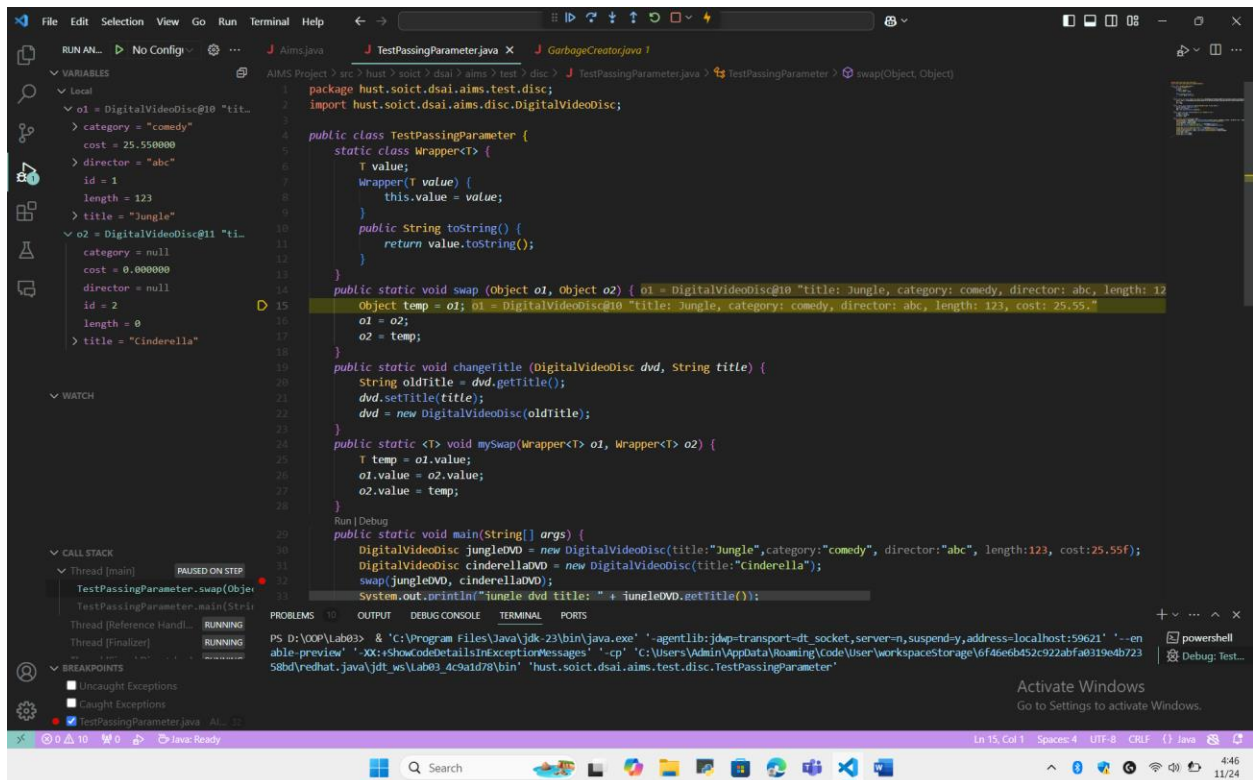


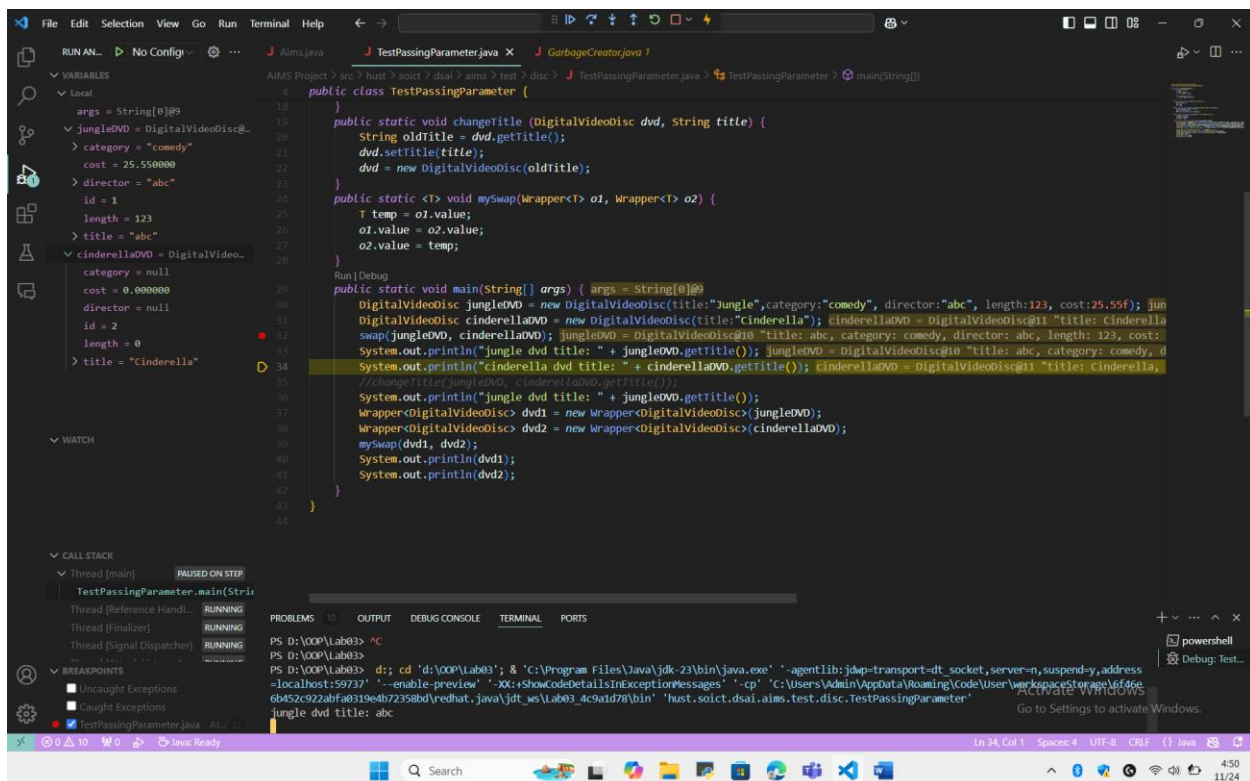
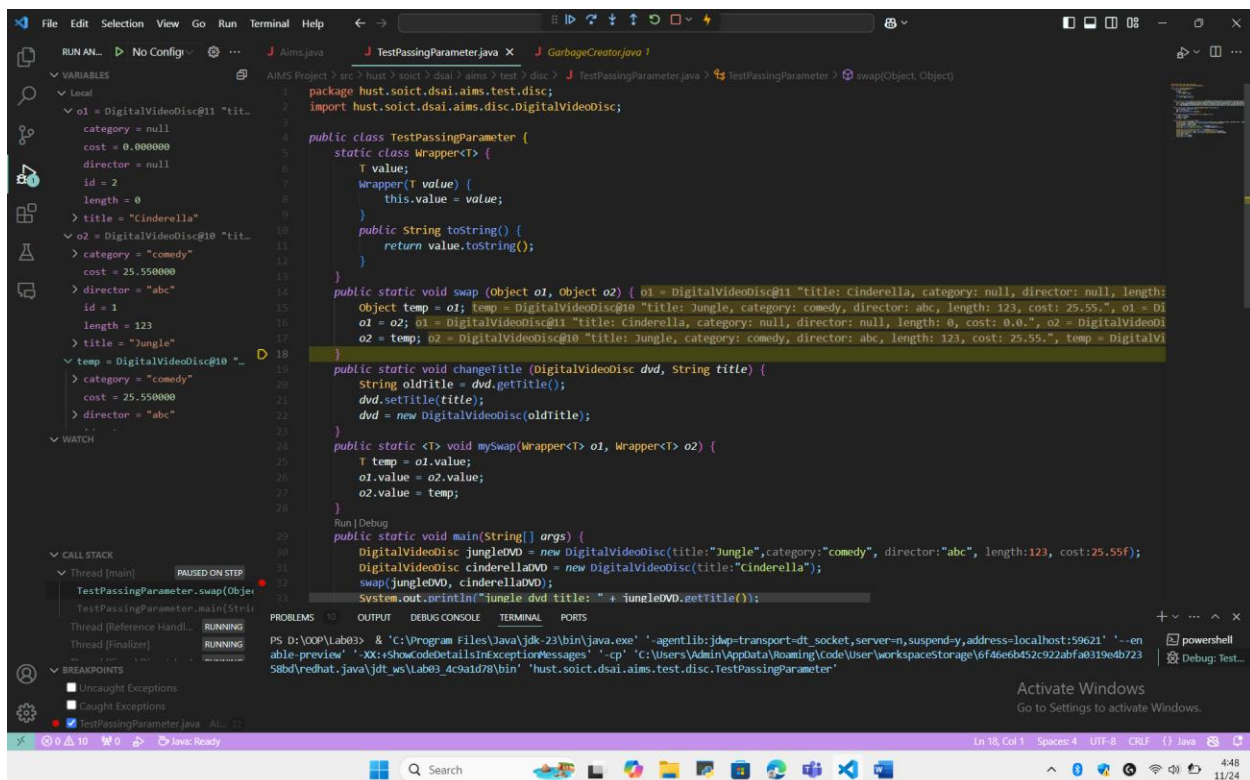
The screenshot shows an IDE with the following code in `TestPassingParameter.java`:

```
4 public class TestPassingParameter {
5     static class Wrapper<T> {
6         T value;
7         Wrapper(T value) {
8             this.value = value;
9         }
10        public String toString() {
11            return value.toString();
12        }
13    }
14    public static void swap (Object o1, Object o2) {
15        Object temp = o1;
16        o1 = o2;
17        o2 = temp;
18    }
19    public static void changeTitle (DigitalVideoDisc dvd, String title) {
20        String oldTitle = dvd.getTitle();
21        dvd.setTitle(title);
22        dvd = new DigitalVideoDisc(oldTitle);
23    }
24    public static <T> void mySwap(Wrapper<T> o1, Wrapper<T> o2) {
25        T temp = o1.value;
26        o1.value = o2.value;
27        o2.value = temp;
28    }
29    Run | Debug
30    public static void main(String[] args) {
31        DigitalVideoDisc jungleDVD = new DigitalVideoDisc(title:"Jungle",category:"comedy", director:"abc", length:123, cost:25.55f);
32        DigitalVideoDisc cinderellaDVD = new DigitalVideoDisc(title:"Cinderella");
33        swap(jungleDVD, cinderellaDVD);
34        System.out.println("jungle dvd title: " + jungleDVD.getTitle());
35        System.out.println("cinderella dvd title: " + cinderellaDVD.getTitle());
36        //changeTitle(jungleDVD, cinderellaDVD.getTitle());
37        System.out.println("jungle dvd title: " + jungleDVD.getTitle());
38        Wrapper<DigitalVideoDisc> dvd1 = new Wrapper<DigitalVideoDisc>(jungleDVD);
39        Wrapper<DigitalVideoDisc> dvd2 = new Wrapper<DigitalVideoDisc>(cinderellaDVD);
40        mySwap(dvd1, dvd2);
41        System.out.println(dvd1);
42        System.out.println(dvd2);
43    }
44 }
```

The IDE interface includes an Explorer on the left showing the project structure, a Run and Debug console at the bottom, and a status bar at the very bottom.

3) Use Debug Run





4) Classifier Member and Instance Member

The screenshot shows an IDE window with the following components:

- Menu Bar:** File, Edit, Selection, View, Go, Run, Terminal, Help.
- Toolbar:** Includes icons for file operations (new, open, save, delete), search, and running.
- EXPLORER Panel:** Displays the project structure under 'LAB03'. The 'DigitalVideoDisc.java' file is selected under the 'disc' folder.
- Editor:** Displays the code for 'DigitalVideoDisc.java'. The code includes package declarations, class declarations, and several methods for managing video discs.
- Bottom Panel:** Includes 'OUTLINE', 'TIMELINE', and 'JAVA PROJECTS' tabs. The 'OUTLINE' tab is active, showing a list of methods in the class.
- Status Bar:** Shows 'Ln 1, Col 1', 'Spaces: 4', 'UTF-8', 'CR/LF', '1 - Java', and the time '2:49 11/24'.

```
package hust.soiict.dsai.aims.disc;

public class DigitalVideoDisc {
    private String title;
    private String category;
    private String director;
    private int length;
    private float cost;
    private static int nbDigitalVideoDisc;
    private int id;

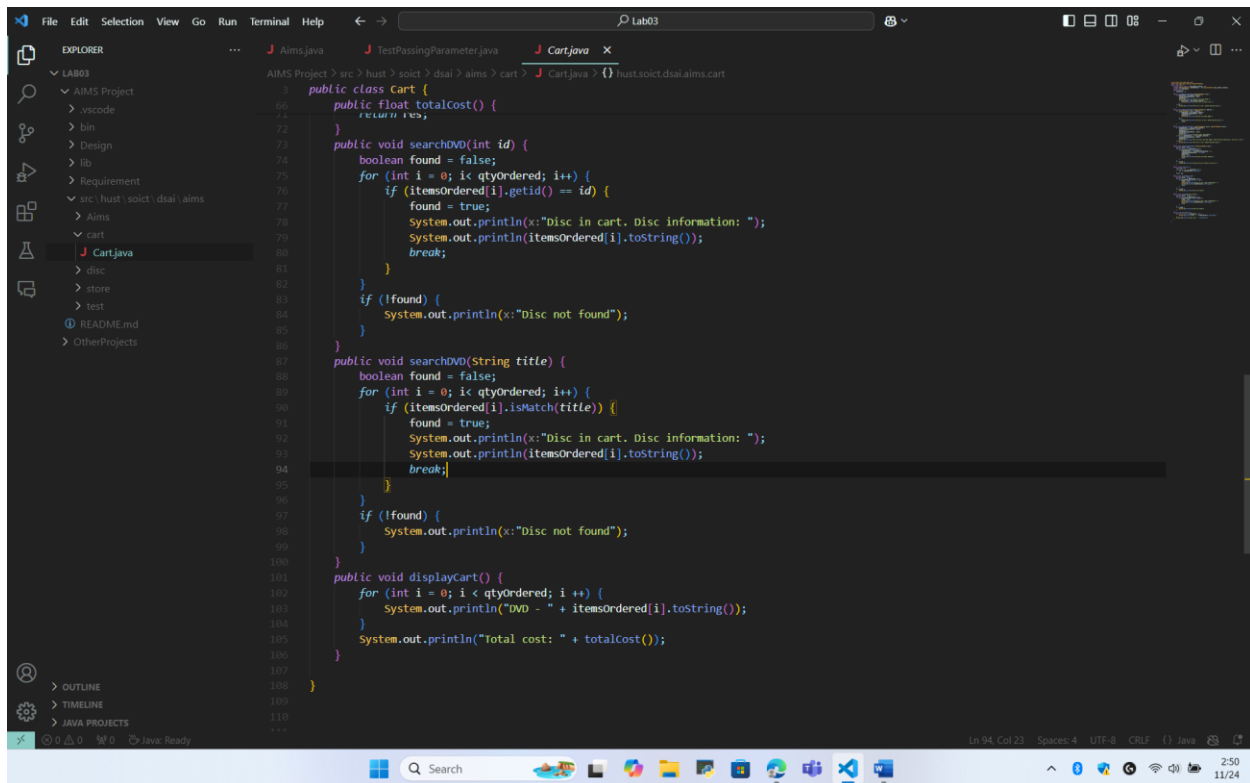
    public String getTitle() { ... }
    public void setTitle(String title) { ... }
    public String getCategory() { ... }
    public void setCategory(String category) { ... }
    public String getDirector() { ... }
    public void setDirector(String director) { ... }
    public int getLength() { ... }
    public void setLength(int length) { ... }
    public float getCost() { ... }
    public void setCost(float cost) { ... }
    public int getId() {
        return id;
    }

    public DigitalVideoDisc(String title) {
        this.title = title;
        nbDigitalVideoDisc ++;
        this.id = nbDigitalVideoDisc;
    }

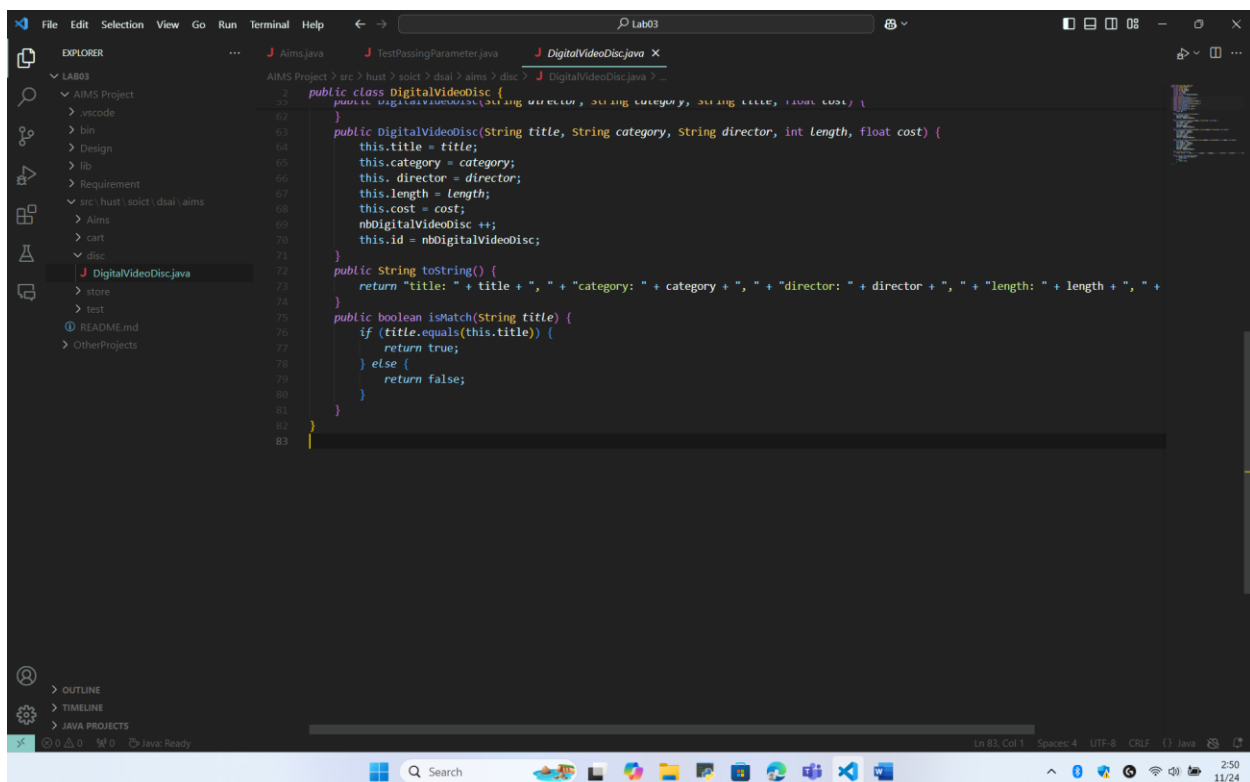
    public DigitalVideoDisc(String category, String title, float cost) {
        this.category = category;
        this.title = title;
        this.cost = cost;
        nbDigitalVideoDisc ++;
        this.id = nbDigitalVideoDisc;
    }

    public DigitalVideoDisc(String director, String category, String title, float cost) {
        this.director = director;
        this.category = category;
        this.title = title;
        this.cost = cost;
        nbDigitalVideoDisc ++;
        this.id = nbDigitalVideoDisc;
    }
}
```

5) Cart class



```
public class Cart {  
    public float totalCost() {  
        return res;  
    }  
    public void searchDVD(int id) {  
        boolean found = false;  
        for (int i = 0; i < qtyOrdered; i++) {  
            if (itemsOrdered[i].getId() == id) {  
                found = true;  
                System.out.println(x:"Disc in cart. Disc information: ");  
                System.out.println(itemsOrdered[i].toString());  
                break;  
            }  
        }  
        if (!found) {  
            System.out.println(x:"Disc not found");  
        }  
    }  
    public void searchDVD(String title) {  
        boolean found = false;  
        for (int i = 0; i < qtyOrdered; i++) {  
            if (itemsOrdered[i].isMatch(title)) {  
                found = true;  
                System.out.println(x:"Disc in cart. Disc information: ");  
                System.out.println(itemsOrdered[i].toString());  
                break;  
            }  
        }  
        if (!found) {  
            System.out.println(x:"Disc not found");  
        }  
    }  
    public void displayCart() {  
        for (int i = 0; i < qtyOrdered; i++) {  
            System.out.println("DVD - " + itemsOrdered[i].toString());  
        }  
        System.out.println("Total cost: " + totalCost());  
    }  
}
```



```
public class DigitalVideoDisc {  
    public DigitalVideoDisc(String director, String category, String title, float cost) {  
    }  
    public DigitalVideoDisc(String title, String category, String director, int length, float cost) {  
        this.title = title;  
        this.category = category;  
        this.director = director;  
        this.length = length;  
        this.cost = cost;  
        nbDigitalVideoDisc++;  
        this.id = nbDigitalVideoDisc;  
    }  
    public String toString() {  
        return "title: " + title + ", " + "category: " + category + ", " + "director: " + director + ", " + "length: " + length + ", " +  
    }  
    public boolean isMatch(String title) {  
        if (title.equals(this.title)) {  
            return true;  
        } else {  
            return false;  
        }  
    }  
}
```

6) Store Class

```
7 public class Store {
8     private List<DigitalVideoDisc> itemsInStore;
9     public Store() {
10         itemsInStore = new ArrayList<>();
11     }
12     public void addDVD(DigitalVideoDisc dvd) {
13         boolean found = false;
14         for (DigitalVideoDisc disc : itemsInStore) {
15             if (disc.equals(dvd)) {
16                 found = true;
17                 break;
18             }
19         }
20         if (found) {
21             System.out.println(x:"The dvd is already added");
22         } else {
23             itemsInStore.add(dvd);
24             System.out.println(x:"The dvd has been added");
25         }
26     }
27     public void removeDVD(DigitalVideoDisc dvd) {
28         boolean found = false;
29         for (DigitalVideoDisc disc : itemsInStore) {
30             if (disc.equals(dvd)) {
31                 found = true;
32                 break;
33             }
34         }
35         if (found) {
36             itemsInStore.remove(dvd);
37             System.out.println(x:"The dvd has been removed");
38         } else {
39             System.out.println(x:"DVD not found");
40         }
41     }
42     public void displayStore() {
43         for (DigitalVideoDisc disc : itemsInStore) {
44             System.out.println(disc.toString());
45         }
46     }
47 }
```

7) StringBuilder, StringBuffer

This screenshot shows the Visual Studio Code editor with the file `ConcatenationInLoops.java` open. The Explorer sidebar on the left shows the project structure for 'LAB03', including folders like 'src' and 'garbage', and files like 'ConcatenationInLoops.java', 'GarbageCreator.java', and 'NoGarbage.java'. The main editor area displays the following Java code:

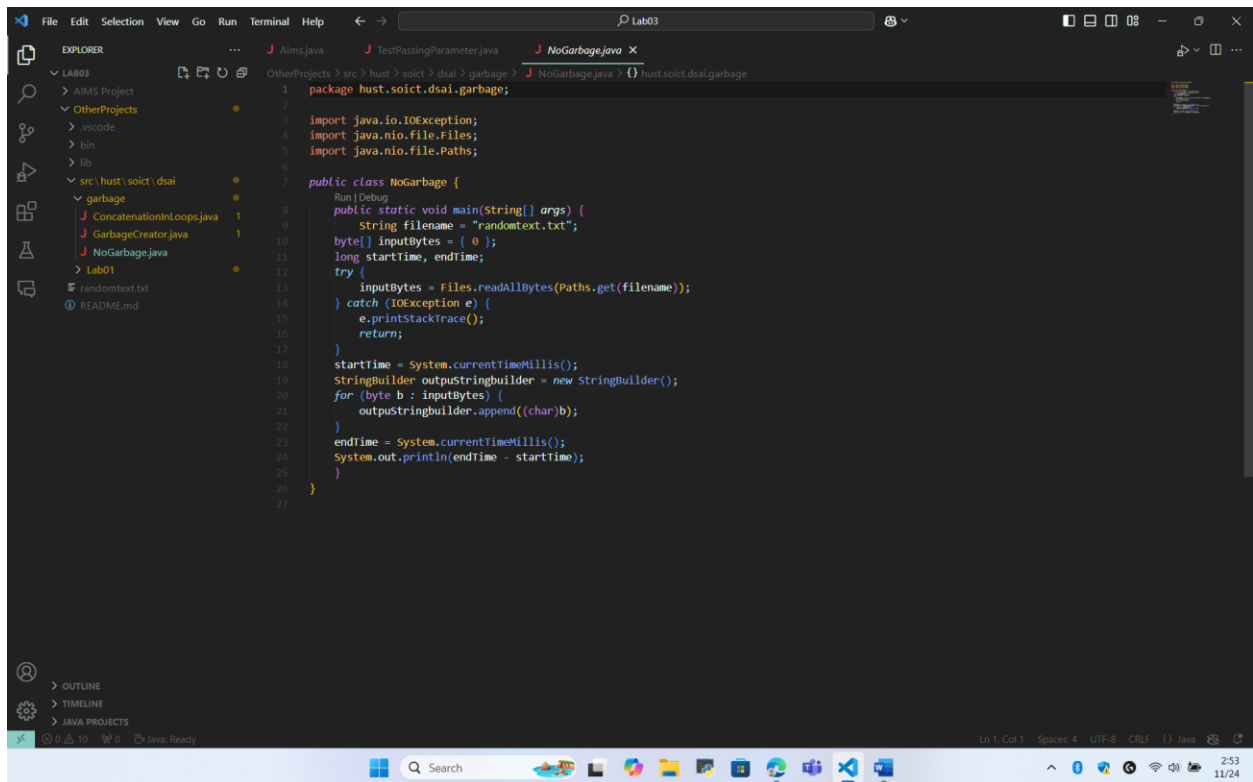
```
1 package hust.soict.dsai.garbage;
2
3 import java.util.Random;
4
5 public class ConcatenationInLoops {
6     public static void main(String[] args) {
7         Random r = new Random(seed:123);
8         long start = System.currentTimeMillis();
9         String s = "";
10        for (int i = 0; i < 65536; i++) {
11            s += r.nextInt(bound:2);
12        }
13        System.out.println(System.currentTimeMillis() - start);
14
15        r = new Random(seed:123);
16        start = System.currentTimeMillis();
17        StringBuilder sb = new StringBuilder();
18        for (int i = 0; i < 65536; i++) {
19            sb.append(r.nextInt(bound:2));
20        }
21        s = sb.toString();
22        System.out.println(System.currentTimeMillis() - start);
23    }
24 }
25
26 }
```

The status bar at the bottom indicates 'Ln 1, Col 1', 'Spaces: 4', 'UTF-8', 'CR/LF', and 'Java'. The system tray at the very bottom shows the time as 2:53 on 11/24.

This screenshot shows the Visual Studio Code editor with the file `GarbageCreator.java` open. The Explorer sidebar on the left shows the project structure for 'LAB03', including folders like 'src' and 'garbage', and files like 'ConcatenationInLoops.java', 'GarbageCreator.java', and 'NoGarbage.java'. The main editor area displays the following Java code:

```
1 package hust.soict.dsai.garbage;
2
3 import java.io.IOException;
4 import java.nio.file.Files;
5 import java.nio.file.Paths;
6
7 public class GarbageCreator {
8     public static void main(String[] args) {
9         String filename = "randomtext.txt";
10        byte[] inputBytes = { 0 };
11        long startTime, endTime;
12        try {
13            inputBytes = Files.readAllBytes(Paths.get(filename));
14        } catch (IOException e) {
15            e.printStackTrace();
16            return;
17        }
18        startTime = System.currentTimeMillis();
19        String outputString = "";
20        for (byte b : inputBytes) {
21            outputString += (char)b;
22        }
23        endTime = System.currentTimeMillis();
24        System.out.println(endTime - startTime);
25    }
26 }
27 }
```

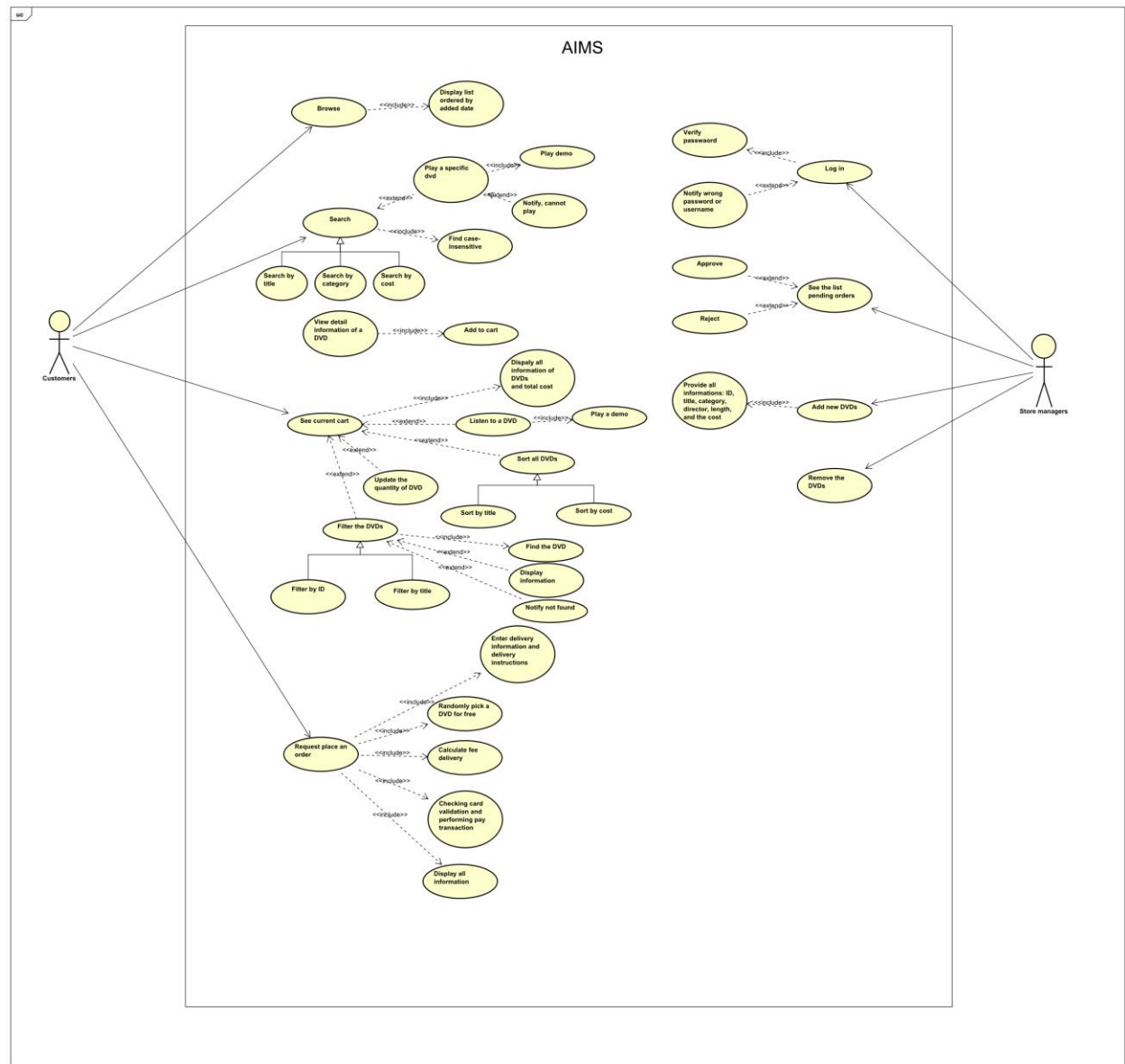
The status bar at the bottom indicates 'Ln 1, Col 1', 'Spaces: 2', 'UTF-8', 'CR/LF', and 'Java'. The system tray at the very bottom shows the time as 2:53 on 11/24.

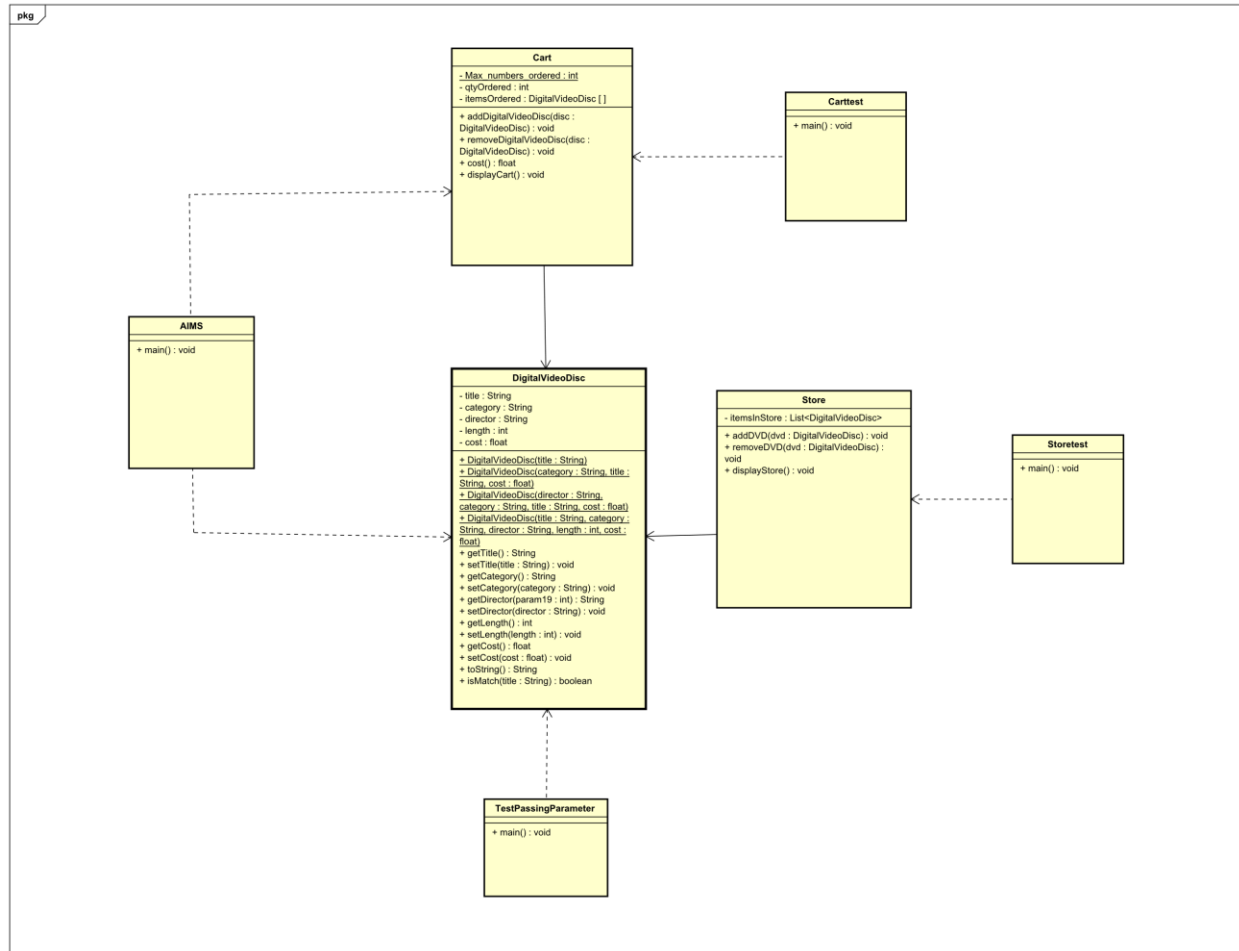


```
1 package hust.soict.dsai.garbage;
2
3 import java.io.IOException;
4 import java.nio.file.Files;
5 import java.nio.file.Paths;
6
7 public class NoGarbage {
8     public static void main(String[] args) {
9         String filename = "randomtext.txt";
10        byte[] inputBytes = { 0 };
11        long startTime, endTime;
12        try {
13            inputBytes = Files.readAllBytes(Paths.get(filename));
14        } catch (IOException e) {
15            e.printStackTrace();
16            return;
17        }
18        startTime = System.currentTimeMillis();
19        StringBuilder outputStringBuilder = new StringBuilder();
20        for (byte b : inputBytes) {
21            outputStringBuilder.append((char)b);
22        }
23        endTime = System.currentTimeMillis();
24        System.out.println(endTime - startTime);
25    }
26
27 }
```

II. Updated diagram

2) Class





III. Answer questions

- 1) After the call of **swap(jungleDVD, cinderellaDVD)** why does the title of these two objects still remain?
- 2) After the call of **changeTitle(jungleDVD, cinderellaDVD.getTitle())** why is the title of the JungleDVD changed?
- 3) **Is JAVA a Pass by Value or a Pass by Reference programming language?**

Java is a pass by value programming language, so after the call of **swap(jungleDVD, cinderellaDVD)**, this method only swaps the copies of the references, not the actual references, hence these two objects still remain. In the second method, the dvd parameter is a reference to the same DigitalVideoDisc object as jungle DVD, hence when you use method setTitle, it change the title of that actual object.

Conclusion, Java is a pass by value programming language.